

Correlation between structural and electrical properties of heavily lithiated boron oxide solid electrolytes

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Abstract

Infrared and impedance measurements have been carried out on high lithium content boron oxide solid electrolytes $x\text{Li}_2\text{O}-\text{B}_2\text{O}_3$, with $x = 3.0\text{--}4.5$, in order to establish the correlation between their structural and electrical properties. The samples are two-phase materials, in which crystalline islands are scattered throughout the vitreous background phase. The evolution of structural and electrical properties with increasing x exhibits a marked differentiation around $x = 4.0$, which is associated with increasing crystallization in the material.
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1. Introduction

Oxide glasses were among the first solid state materials that were studied for their high ionic conductivity. Vitreous lithium borates [1,2] in particular, are a class of ion-conducting glasses that have been considered as potential solid electrolytes in a variety of applications, such as solid state lithium-ion batteries [3], microbatteries [4,5] and sensors. Boron oxide (B_2O_3) is an excellent network former, readily forming glasses when melted and quenched. The addition of lithium oxide (Li_2O) into the borate glass introduces mobile lithium ions in the resulting vitreous $x\text{Li}_2\text{O}-\text{B}_2\text{O}_3$ solid solution, giving rise to ionic conductivity [1,2]. Higher lithium content implies greater ionic conductivity. Formerly, the maximum molar mixing proportion of Li_2O and B_2O_3 , for which it was achieved to prepare a purely vitreous product by melting and subsequent rapid quenching, was 70% Li_2O , corresponding to $x = 2.33$ [1,2]. Higher lithium content boron oxide solid electrolytes, prepared by

quenching, exhibit some crystallinity, the extent of which depends on the cooling rate. The presence of the crystalline phase, in the form of isolated islands scattered within the vitreous phase, hinders further increase in ionic conductivity with increasing lithium content [6]. Very recently, the glass forming range of lithium borates has been extended to 83% mol Li_2O , corresponding to $x = 5.0$, using a solution method [7].

To our knowledge, despite the fact that the binary system $\text{Li}_2\text{O}-\text{B}_2\text{O}_3$ is well known, a study of the structural and/or electrical properties of high lithium content borates, which are essentially two-phase systems, has not been presented in the literature. We have recently proposed an equivalent circuit model to account for the impedance properties of two-phase materials that conform to the particular micrometer-scale structure [6] of high lithium content borates [8]. The results turned out by the application of the model indicated a marked differentiation between two groups of samples, up to $x = 3.75$ on one hand, and greater than $x = 4.0$ on the other. This differentiation has been tentatively attributed to the peculiar structure of the samples [8].

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In this paper, we present infrared (IR) and impedance measurements of high lithium content boron oxide solid electrolytes, in an attempt to correlate their structural and electrical properties. High concentration of Li has been proved to increase considerably the ionic conductivity of the materials [6,8]. Similar work, to account for the structure of such two-phase materials, or the origin of the differentiation in the samples' electrical properties, has not been reported yet in the literature.

2. Experimental

The lithium boron oxide glass samples $x\text{Li}_2\text{O}-\text{B}_2\text{O}_3$, with $x = 3.0, 3.25, 3.5, 3.75, 4.0$ and 4.5 , were prepared by melting appropriate quantities of Li_2CO_3 and B_2O_3 in a furnace. Carbon is removed from the melt as CO_2 , produced by the thermal decomposition of Li_2CO_3 to Li_2O . Details have been given elsewhere [6]. The samples were shaped in the form of thin sheets by pressing drops of the melt between stainless steel plates. Up to $x = 3.75$, this quenching procedure yields transparent samples which, nevertheless, contain small white spots in their volume, an indication of the formation of crystalline regions in the vitreous bulk background. The presence of crystalline islands, being gradually larger in size and concentration, becomes more pronounced as the lithium content in the samples increases, until, for $x = 4.0$ and 4.5 , the background phase takes a milky semi-transparent appearance and the crystalline islands are densely scattered within. The classification of the samples into two categories according to their micrometer-scale structure, $x = 3.0, 3.25, 3.5$ and 3.75 on one hand, and $x = 4.0$ and 4.5 on the other, has been demonstrated in previous reports [6,8] by employing scanning electron microscopy (SEM).

Gold electrodes were developed on both sides of the samples by thermal evaporation, in order for electrical measurements to be performed. Impedance spectra were recorded as a function of temperature in the range from ambient temperature to 220°C . The frequency range scanned was 10 mHz to 1 MHz. During measurement, the samples were kept in vacuum of the order of 1 Pa and at very stable temperature. The error in the temperature value corresponding to each impedance spectrum, caused by temperature fluctuation during measurement of the spectrum, was in all cases not greater than 0.2%. IR spectra of the samples, embedded in KBr pellet, were recorded at 2 cm^{-1} resolution by a Fourier transform infrared (FTIR) spectrometer equipped with a deuterated triglycine sulfate (DTGS) detector. For the absorbance spectra the instrument background was applied as reference.

3. Results

The impedance spectra measured from the samples typically show two electrical relaxation time constants. The Nyquist plots consist of two consecutive semi-circular arcs and a low-frequency electrode polarization straight line.

An example, obtained from the $x = 3.75$ sample at 97°C , is shown in Fig. 1. The fitting circuit that was used for data analysis is shown in Fig. 2 [8]. It turns out that the previously mentioned differentiation of high lithium content borate samples with respect to their micrometer-scale structure [6] is followed in the properties of ionic conductivity as well. More specifically, the temperature dependence of resistors R_1 and R_2 of the fitting circuit is found to differentiate between the two categories of samples. The Arrhenius plot of $G_1 = R_1^{-1}$ is always linear. On one hand, for $x = 3.0, 3.25, 3.5$ and 3.75 , the Arrhenius plot of $G_2 = R_2^{-1}$ is non-linear, tending asymptotically to become linear in the low- and high-temperature limits. On the other hand, for $x = 4.0$ and 4.5 both the G_1 and G_2 plots turn out to be linear. The values of R_1 and R_2 yielded by fitting were fairly accurate, with an error of less than 1% in all cases, with the exception of a few high-temperature points, in which the error of R_2 was larger, but not exceeded 5% [8].

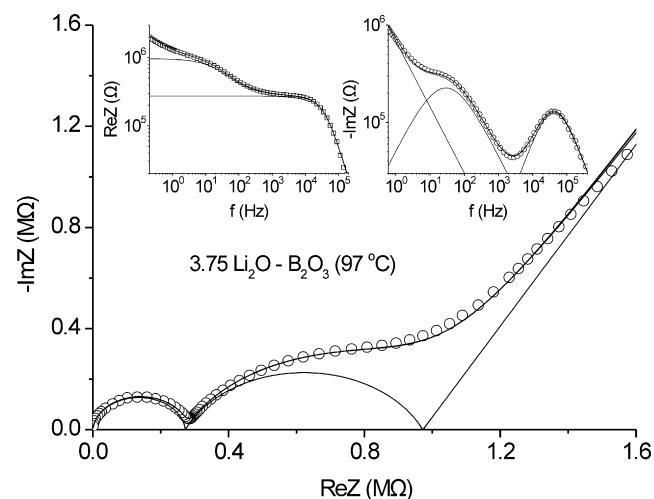


Fig. 1. Impedance spectrum in Nyquist representation of the $x = 3.75$ sample at 97°C , along with its spectral components. In the insert plot are the corresponding Bode plots of the real and imaginary parts of the impedance.

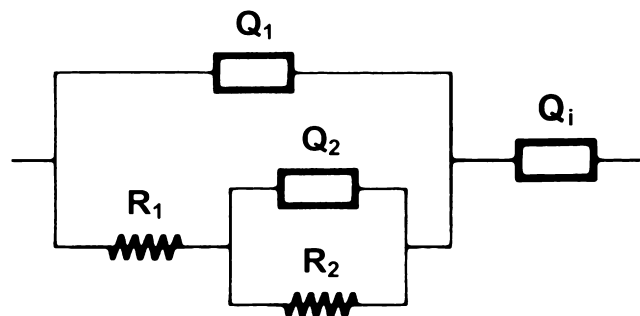


Fig. 2. The fitting circuit used for analysis of impedance spectra measured from two-phase materials. Two pairs of constant phase elements and resistors, (Q_1, R_1) and (Q_2, R_2) , model the material electrical properties. The third constant phase element Q_i models the electrode polarization capacitance.

As an example, the Arrhenius plots for $x = 3.75$ and 4.0 are shown in Fig. 3. For the sake of comparison between different samples, the calculated circuit conductances G_1 and G_2 were converted to the corresponding conductivities σ_1 and σ_2 , by taking into account the thickness of each sample and the macroscopic surface area of the electrodes. The σ_1 Arrhenius plots are compared in Fig. 3(a), and the σ_2 plots are compared in Fig. 3(b). The values of ionic conductivity activation energy and pre-exponential factor, associated with the process of ionic conduction in the two phases present in the samples, namely the vitreous and the crystalline, were calculated by fitting the Arrhenius equation to the conductivity data. The drop in the calculated activation energy of the crystalline phase, from $E_C = 1.925 \pm 0.158$ eV for $x = 3.75$ to $E_C = 0.716 \pm 0.012$ eV for $x = 4.0$ is remarkable. The calculated activation energy of the vitreous phase increases from $E_V = 0.582 \pm 0.003$ eV for $x = 3.75$ to $E_V = 0.679 \pm 0.010$ eV

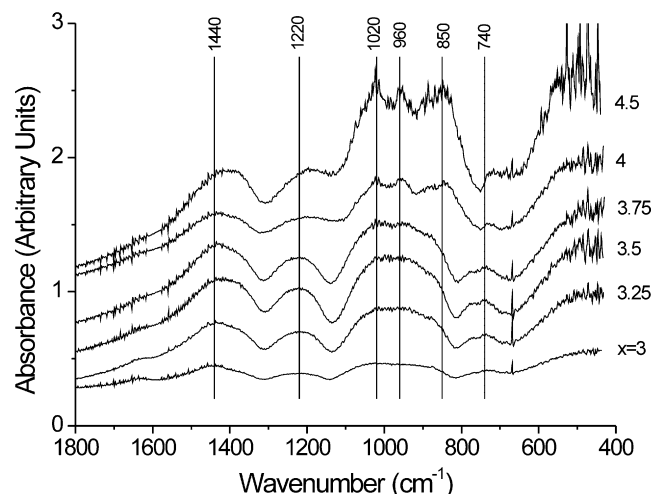


Fig. 4. Infrared absorption spectra of the high lithium content borate samples.

for $x = 4.0$. In the transition from $x = 3.75$ to $x = 4.0$, an equally remarkable drop by 15 orders of magnitude is observed in the pre-exponential factor of the crystalline phase, from $F_C = 7.33 \times 10^{18}$ S/cm (with $\log F_C = 18.865 \pm 0.660$) to $F_C = 9.25 \times 10^3$ S/cm (with $\log F_C = 3.966 \pm 0.153$). At the same time, the pre-exponential factor of the vitreous phase increases from $F_V = 1.52 \times 10^2$ S/cm (with $\log F_V = 2.182 \pm 0.042$) to $F_V = 1.45 \times 10^3$ S/cm (with $\log F_V = 3.162 \pm 0.130$).

The IR absorption spectra of the samples, with x ranging from 3.0 to 4.5, are shown in Fig. 4. The most striking feature of the spectra is the presence of broad noisy peaks located at the low-frequency end, in the region with wavenumbers lower than 600 cm^{-1} . Pure Li_2O is known to exhibit a broad absorption band centered at $450\text{--}500 \text{ cm}^{-1}$. These peaks can, therefore, be attributed to the presence of crystalline Li_2O in the samples. The other peaks in the IR spectra originate from B–O bond vibrations of the elementary units making up the complex atomic groups present in the samples. Discrete peaks are located at 740 cm^{-1} , 1220 cm^{-1} and 1440 cm^{-1} . Most importantly, a very broad absorption band can be seen in the region $800\text{--}1100 \text{ cm}^{-1}$. This particular band, in the case of the highest lithium content samples $x = 4.0$ and 4.5 , presents a fine structure, splitting into discrete peaks, the strongest of which appear at about 850 , 960 and 1020 cm^{-1} . The differentiation of these samples with respect to the lower lithium content samples is immediately evident, indicating a corresponding marked alteration in the material structure.

4. Discussion

The formation of two arcs in impedance plots is characteristic of two-phase systems, and each arc is associated with one of the present phases. We have proposed an equivalent circuit model to describe the electrical properties of two-phase systems, particularly the non-Arrhenius tem-

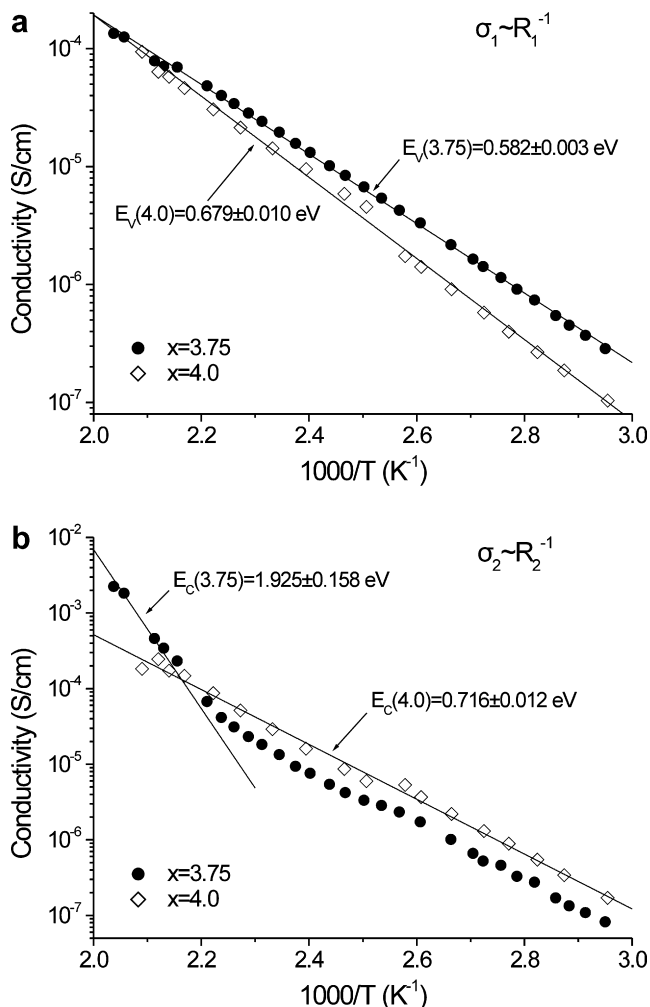


Fig. 3. Comparison of conductivity Arrhenius plots between the $x = 3.75$ and 4.0 lithium borate samples, σ_1 (a) and σ_2 (b), which were calculated from $G_1 = R_1^{-1}$ and $G_2 = R_2^{-1}$, respectively, by considering sample geometry. The values of R_1 and R_2 are yielded by fitting the circuit of Fig. 2 to impedance spectra.

perature dependence of the components of ionic conductivity, as deduced from conventional analysis of the semi-circular arcs in Nyquist plots [8]. The model will be briefly summarized here.

The two-phase material is supposed to be composed of a continuous vitreous background phase, in which isolated grains of another, crystalline, phase are dispersed. The ionic current through this system is divided into two components, the first passing through the vitreous phase alone, and the second passing through parts of both the vitreous and crystalline phases, and which is partly blocked at the interfaces separating the grains from the background. The equivalent circuit model reduces to the fitting circuit of Fig. 2, which is used for data analysis [8], since impedance spectra fitting cannot be carried out by directly employing the model due to over-parameterization. The fitting circuit parameters are related to the electrical properties of the constituent material phases by mathematical formulae. The temperature dependence of resistors R_1 and R_2 , and their relationship to the ionic conductivity activation energy of the two phases is of particular interest. The most important implication is that resistance R_2 should vary with temperature in a non-Arrhenius manner [8].

The model successfully explains the shape of Arrhenius plots resulting from the high lithium content borates under study, such as those of Fig. 3. According to the model, the Arrhenius plot of $G_1 = R_1^{-1}$ should turn out to be linear, the slope of which depends on the vitreous phase activation energy E_V . On the other hand, the Arrhenius plot of $G_2 = R_2^{-1}$ should, in general, turn out to be non-linear. The slope of the low-temperature asymptote of the G_2 plot is directly related to the vitreous phase activation energy E_V , while the slope of the high-temperature asymptote is directly related to the crystalline phase activation energy E_C [8]. However, as is demonstrated by Fig. 3, for $x = 4.0$ and higher both the G_1 and G_2 plots are indeed linear. We have attributed this differentiation to the extremely dense distribution of crystalline islands in the vitreous phase, with direct consequence in the elimination of the current component that passes through the vitreous phase alone. In this case, the model is in one-to-one correspondence to the fitting circuit [8].

The differences in the calculated activation energies and pre-exponential factors of both the phases, especially the crystalline, that can be observed between the two differentiated groups of samples, have to be traced to corresponding differences in their composition and structure. In particular, the substantial difference regarding the values of pre-exponential factor has been attributed largely to the geometrical effect that results from the different occupation of the total sample volume by the crystalline phase [8]. Clearly, the pre-exponential factors represent not only the electrical properties of the two constituent phases, but also the geometry of their distribution in the samples, which in any case is extremely complex. However, as has already been mentioned above, up to $x = 3.75$ the occupation of

the vitreous background phase from the isolated islands of the crystalline phase is relatively limited, whereas from $x = 4.0$ the crystalline islands within the vitreous phase become extremely concentrated [8]. This differentiation, in the micrometer scale, of the geometry of the two-phase system in the samples is the origin of the difference in the pre-exponential factors associated with the crystalline phase [8].

A detailed investigation and interpretation of the electrical properties of high lithium content borates has been published in previous reports [6,8]. The main aim of the present work is to illustrate the observed differentiation between the two groups of samples, especially in connection with the ionic conductivity activation energy, by investigating the sample structure in the atomic scale. Infrared spectroscopy was employed to this purpose.

The structure of low lithium content borate glasses is known from the literature [1,2,9]. The elementary units, out of which the vitreous network is composed, are triangular BO_3 and tetrahedral BO_4 . These units exist either as discrete groups, or as part of more complex ones. Pure vitreous B_2O_3 contains only triangular BO_3 units, mostly as parts of hexagonal boroxol rings B_3O_6 . With the addition of Li_2O , each added oxygen atom results in the transformation of one or two three-coordinated boron atoms into four-coordinated ones, depending on whether the added atom is incorporated into the network as non-bridging oxygen (NBO) or bridging oxygen (BO), respectively. The anionic groups involving only BO, in order of increasing complexity, are loose BO_4^- tetrahedra, triborate B_3O_5^- , pentaborate B_5O_8^- , tetraborate $\text{B}_4\text{O}_{13}^{2-}$ and diborate $\text{B}_2\text{O}_7^{2-}$. The essential difference between diborate and all the other groups is that the former originates from deformed boroxol rings, two boron atoms of which have been transformed from three- to four-coordinated, while the latter originate from boroxol rings with only one transformed boron atom. For this reason, diborate presents different IR absorption peaks resulting from B–O bond stretching vibrations of tetrahedral BO_4 units, with respect to the other groups. The groups involving NBO are metaborate BO_2^- , orthoborate BO_3^{3-} and pyroborate $\text{B}_2\text{O}_5^{4-}$ [10,11]. The last two are independent anions, containing no BO at all. As the lithium content of borate glasses increases, the vitreous network is gradually depolymerized and the structural groups involving only BO give way to other groups involving NBO, until, roughly for $x > 1$ according to the literature [1,2], the glass is thought to be composed mostly of orthoborate and pyroborate anions.

The broad noisy peaks in the IR spectra of Fig. 4, located in the region beyond 600 cm^{-1} , were attributed to the presence of crystalline Li_2O . The increasing peak intensity with increasing lithium content is in agreement with the larger crystallite density observed in the samples, evident from direct optical inspection and demonstrated by SEM measurements [6]. The crystalline islands scattered throughout the volume of all the samples are hence suggested to be associated mainly with Li_2O .

The rest of the peaks in the IR spectra are attributed to the vibrations of B–O bonds of the elementary units that constitute the complex atomic groups present in the glass structure. Despite the fact that a detailed interpretation of the IR spectra of borate glasses has known to be not an easy task, it is generally accepted that absorption peaks in the region $850\text{--}1100\text{ cm}^{-1}$ originate from B–O bond stretching of tetrahedral BO_4 units, while peaks in the region $1200\text{--}1450\text{ cm}^{-1}$ originate from B–O bond stretching of trigonal BO_3 units [2,12]. The small absorption peak at 740 cm^{-1} is assigned to bending vibrations of B–O–B bonds, linking various elementary units of the vitreous network. The slight decrease of its intensity with increasing lithium content seems to be consistent with the gradual elimination of the covalent chains and the appearance of discrete borate anions in the glass structure, as is stated in the literature [2,9]. The peak at 1220 cm^{-1} can similarly be assigned to stretching vibrations of B–O bonds, involving BO atoms participating in B–O–B linkages between different units. Since it is widely accepted that no boroxol rings are present in high lithium content borate glasses, this absorption band should originate from triborate, tetraborate and pentaborate groups [9,12]. This peak can also be attributed to asymmetric stretching vibrations of B-O^- bonds in pyroborate $\text{B}_2\text{O}_5^{4-}$ and orthoborate BO_3^{3-} units [2,13]. On this basis, the absorption band at about 1220 cm^{-1} indicates the presence of all the aforementioned groups in the samples. In the literature [1,2] this particular band has been interpreted as originating mainly from orthoborate and pyroborate groups, since the breakdown of the vitreous network under high Li_2O doping excludes the existence of large covalently linked borate groups. However, the obvious high crystallinity of the samples suggests that the presence of crystalline triborate, tetraborate and pentaborate lithium compounds is not impossible. Further evidence is provided from other features of the IR spectra, as is explained further below.

The broad absorption band centered at 1440 cm^{-1} is attributed to B–O bond stretching of trigonal BO_3 units, also indicating the presence of the orthoborate and mainly the pyroborate group in the glass. The gradual increase of the intensity of this band with increasing lithium content seems to be in agreement with the widely accepted statement that discrete borate anions dominate the structural units of high lithium content glasses.

The most remarkable feature of the IR spectra is the broad absorption band in the region $800\text{--}1100\text{ cm}^{-1}$, which for $x = 4.0$ and 4.5 presents discrete peaks at 850 , 960 and 1020 cm^{-1} , revealing the differentiation of these samples in comparison to the lower lithium content samples. This differentiation indicates a corresponding variation in the material structure. The different structure is also demonstrated by SEM [6], and is as well reflected in the electrical ionic conduction properties, as has already been stated. Crystalline tetraborate compounds show strong absorption bands at 880 and 1020 cm^{-1} , while crystalline diborate at

970 cm^{-1} [9]. Thus, the 850 and 1020 cm^{-1} peaks can be attributed to BO_4 vibrations of triborate, tetraborate and pentaborate groups. The analogous vibrations of BO_4 units belonging to diborate groups give rise to absorption in the $900\text{--}1000\text{ cm}^{-1}$ region, so we may assume that the peak at 960 cm^{-1} is due to diborate.

It is certain that all the lithium borate samples, as prepared, are extremely inhomogeneous. This is particularly true for the very high lithium content ones $x = 4.0$ and 4.5 , in which the formation of not only the scattered crystalline islands, but also the continuous background phase, depends on the local cooling rate of the melt during preparation. The binary system $\text{Li}_2\text{O-B}_2\text{O}_3$ is known to form vitreous solid solutions in a wide range of compositions [1], as well as nine crystalline single phases [14,15]. It is, therefore, possible that several different crystalline phases are simultaneously present in the background phase. We may further assume that, in addition to Li_2O , several crystalline lithium borates are present in the scattered crystalline islands, too. Local variations in the stoichiometric composition of vitreous domains of the background phase are also possible. It may be concluded that the differentiation of the samples into two categories, a fact that is supported by impedance, SEM and IR measurements, results from crystallization of the continuous background phase, possibly to a large extent. However, the details of the structure are far from clear, not only in the case of inhomogeneous samples, but also for purely vitreous high alkali content borate glasses, as has been reported recently by other authors [7].

5. Conclusions

High lithium content boron oxide solid electrolytes $x\text{Li}_2\text{O-B}_2\text{O}_3$, with $x = 3.0\text{--}4.5$, prepared by quenching, are inhomogeneous two-phase systems, in which the crystalline phase is dispersed within the vitreous background phase. The samples are divided into two categories, of relatively lower (up to $x = 3.75$) and of relatively higher (greater than $x = 4.0$) lithium content. The differentiation in the samples' structural and electrical properties is supported by SEM, impedance and IR measurements. The dispersed islands of the crystalline phase in the vitreous background phase are mainly associated with Li_2O . In the samples of relatively higher lithium content, $x = 4.0$ and 4.5 , IR measurements indicate the presence of several borate anionic groups. We suggest that the differentiation between the two groups of samples is due to a, possibly large, extent of crystallization in both the background and the dispersed phase. The possible simultaneous existence of several crystalline lithium borate compounds is not excluded.

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References

- [1] C. Julien, G.A. Nazri, *Solid State Batteries – Materials Design and Optimization*, Kluwer Academic Publishers, Boston, 1994.
- [2] M. Balkanski (Ed.), *Microionics – Solid State Integrable Microbatteries*, North Holland, Amsterdam, 1991.
- [3] M. Duclot, J.L. Souquet, *J. Power Sources* 97&98 (2001) 610.
- [4] P. Birke, W. Weppner, *Electrochim. Acta* 42 (1997) 3375.
- [5] C.H. Chen, E.M. Kelder, J. Schoonman, *J. Power Sources* 68 (1997) 377.
- [6] E.E. Horopanitis, G. Perentzis, E. Pavlidou, L. Papadimitriou, *Ionics* 9 (2003) 88.
- [7] J. Banerjee, G. Ongie, J. Harder, T. Edwards, Ch. Larson, S. Sutton, A. Moeller, A. Basu, M. Affatigato, S. Feller, M. Kodama, P.M. Aguiar, S. Kroeker, *J. Non-Cryst. Solids* 352 (2006) 674.
- [8] E.E. Horopanitis, G. Perentzis, L. Papadimitriou, *J. Solid State Electrochem.* 11 (2007) 1171 and 1183.
- [9] J. Krogh-Moe, *J. Phys. Chem. Glasses* 6 (1965) 46.
- [10] E.I. Kamitsos, G.D. Chryssikos, *J. Mol. Struct.* 247 (1991) 1.
- [11] B.V.R. Chowdari, Zh. Rong, *Solid State Ionics* 78 (1995) 133.
- [12] E.I. Kamitsos, M.A. Karakassides, G.D. Chryssikos, *J. Phys. Chem.* 91 (1987) 1073.
- [13] E.I. Kamitsos, M.A. Karakassides, G.D. Chryssikos, *Phys. Chem. Glasses* 28 (1987) 203.
- [14] M.D. Mathews, A.K. Tyagi, P.N. Moorthy, *Thermochim. Acta* 320 (1998) 89.
- [15] A.B. Meshalkin, A.B. Kaplun, *J. Crystal Growth* 275 (2005) e115.